

**Q.1** A manager is attempting to put together an aggregate plan for the coming nine months. She has obtained a forecast of expected demand for the planning horizon. The plan must deal with highly seasonal demand; demand is relatively high in periods 3 and 4 and again in period 8, as can be seen from the following forecasts:

| Period | Forecast |
|--------|----------|
| 1      | 190      |
| 2      | 230      |
| 3      | 260      |
| 4      | 280      |
| 5      | 210      |
| 6      | 170      |
| 7      | 160      |
| 8      | 260      |
| 9      | 180      |

The department now has 20 full-time employees, each of whom produces 10 units of output per period at a cost of \$6 per unit. Beginning inventory for period 1 is zero. Inventory carrying cost is \$5 per unit per period, and backlog cost is \$10 per unit per period.

- Will the current work force be able to handle the forecast demand?
- Determine the total cost of the plan, including production, inventory, and backorder costs.

**Q.2** Determine the total cost for this plan given the following forecast:

| Month | Forecast |
|-------|----------|
| 1     | 380      |
| 2     | 400      |
| 3     | 420      |
| 4     | 440      |
| 5     | 460      |
| 6     | 480      |

Use steady regular output of 400 units per month, use overtime as needed for up to 40 units per month, and use subcontracting to make up any needed output to match the forecast. Unit costs are:

Regular output = \$25 Overtime

= \$40 Subcontract = \$60

Average Balance Inventory = \$15